Deep Neural Networks for Doubly Robust Estimation with Nonprobability Survey Samples

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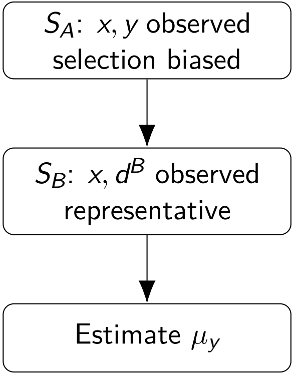
Motivation: integrating two imperfect data sources

- **Nonprobability sample** S_A: rich outcome y, but unknown selection mechanism.
- **Probability sample** S_B: design weights and representative covariates x, but often no y.
- Goal: estimate the finite-population mean

$$\mu_y = \frac{1}{N}\sum_{i=1}^{N} y_i.$$



Formal setup and notation

- Finite population $U = \{1, \ldots, N\}$; target parameter: $\mu_y = \frac{1}{N}\sum_{i=1}^{N} y_i$.
- **Nonprobability sample** S_A, size n_A: observe (x_i, y_i). Selection indicator $R_i = I(i \in S_A)$, with $\pi_i^A = P_q(R_i = 1 \mid x_i, y_i)$.
- **Probability sample** S_B, size n_B: observe (x_i, d_i^B), where $d_i^B = 1/\pi_i^B$ and $\pi_i^B = P(i \in S_B)$ are the first-order inclusion probabilities.
- **Ignorability assumption**: $\pi_i^A = P(R_i = 1 \mid x_i)$ — selection depends only on observed x, not on y.

Three classical strategies

1. Model the outcome: $m(x_i) = E(y_i \mid x_i) \rightarrow$ predict y for units in S_B.
2. Model the selection: $\pi^A(x) = P(R = 1 \mid x) \rightarrow$ inversely weight units in S_A.
3. Combine both models (doubly robust).

Method 1: Regression / mass imputation (REG)

Procedure

1. Fit outcome model $m(x_i) = E(y_i \mid x_i)$ using S_A (e.g., OLS).
2. Predict $\hat{y}_i = m(x_i, \hat{\beta})$ for every unit in S_B.
3. Estimate the population mean by

$$\hat{\mu}_{REG} = \frac{1}{\hat{N}_B} \sum_{i \in S_B} d_i^B \, \hat{y}_i, \qquad \hat{N}_B = \sum_{i \in S_B} d_i^B.$$

Critical assumption

$m(x)$ is correctly specified. If $E(y \mid x)$ contains interactions, thresholds, or nonlinearities and a linear model is used, $\hat{\mu}_{REG}$ is inconsistent — no amount of design weighting can salvage a misspecified outcome model.

Method 2: Inverse probability weighting (IPW)

Procedure

1. Pool S_A and S_B; define $R_i = I(i \in S_A)$.
2. Fit propensity model $\pi^A(x) = P(R = 1 \mid x)$, typically logistic: $\text{logit}[\pi^A(x)] = \alpha_0 + x^\top \alpha$.
3. Weight each unit in S_A inversely to its selection probability:

$$\hat{\mu}_{IPW} = \frac{1}{\hat{N}_A} \sum_{i \in S_A} \frac{1}{\hat{\pi}_i^A} y_i, \quad \hat{N}_A = \sum_{i \in S_A} \frac{1}{\hat{\pi}_i^A}.$$

Critical assumption

$\pi^A(x)$ follows the assumed parametric form (e.g., linear logit). If the true selection mechanism involves **nonlinearities, interactions, or unknown functional forms**, the logistic model is misspecified and IPW is inconsistent.

Method 3: Doubly robust (DR) estimation

Formula

$$\hat{\mu}_{DR} = \frac{1}{\hat{N}_A} \sum_{i \in S_A} \frac{1}{\hat{\pi}_i^A} \{y_i - m(x_i, \hat{\beta})\} + \frac{1}{\hat{N}_B} \sum_{i \in S_B} \frac{1}{\pi_i^B} m(x_i, \hat{\beta}).$$

Double robustness property

$\hat{\mu}_{DR}$ is consistent if **either** the outcome model $m(x)$ **or** the propensity model $\pi^A(x)$ is correctly specified — not necessarily both. This is a genuine improvement over REG and IPW alone.

Remaining vulnerability

When **both** working models are misspecified — e.g., the outcome regression is linear but $E(y \mid x)$ is nonlinear, *and* the propensity is logistic but the true $\pi^A(x)$ is nonlinear — the DR estimator can also be inconsistent.

Parametric double robustness $\neq$ nonparametric protection.

Key idea of the paper

The paper replaces the parametric propensity-score model by a DNN model:

$$\log \frac{\pi_i^A}{1 - \pi_i^A} = g_0(x_i),$$

where g_0 is an unknown nonparametric function.

$$\boxed{\text{Covariates } x} \longrightarrow \boxed{\text{DNN } \hat{g}(x)} \longrightarrow \boxed{\hat{\pi}^A(x)} \longrightarrow \boxed{\text{DIPW / DDR}}$$

The DNN is trained using information from both the nonprobability sample and the reference probability sample.

Why deep neural networks?

- **Universal approximation**: multilayer feedforward networks can approximate broad classes of continuous functions on compact sets.
- **Curse of dimensionality**: under smoothness and structural (e.g. composite Hölder) assumptions, sparse DNNs can achieve the optimal minimax rate without suffering from the curse of dimensionality.
- **Scalability**: compared with classical nonparametric methods (e.g. kernel smoothing), DNNs scale better in moderate-to-high dimensions when the target function has suitable structure.
- **Nonlinear selection**: participation mechanisms in nonprobability samples often involve unknown interactions and nonlinearities — precisely the setting where DNNs excel.

Key insight

The parametric logit model $\text{logit}[\pi^A(x)] = \alpha_0 + x^\top \alpha$ is a **special case** of the DNN model. The DNN relaxes the linear-in-parameters restriction and lets the data determine the functional form of the selection mechanism.

Pseudo-likelihood: derivation

Step 1: Population log-likelihood (infeasible)

If x_i were observed for all $i \in U$, the log-likelihood for the participation indicators R_i would be

$$\ell(g) = \sum_{i=1}^{N} \left[R_i \log \pi_i^A + (1 - R_i) \log(1 - \pi_i^A) \right] = \sum_{i \in S_A} \log \frac{\pi_i^A}{1 - \pi_i^A} + \sum_{i=1}^{N} \log(1 - \pi_i^A).$$

Step 2: Replace population total by HT estimator

Since x_i is unavailable outside $S_A \cup S_B$, the population sum $\sum_{i=1}^{N} \log(1 - \pi_i^A)$ cannot be computed. The **Horvitz–Thompson estimator** uses the probability sample S_B:

$$\sum_{i=1}^{N} \log(1 - \pi_i^A) \approx \sum_{i \in S_B} d_i^B \log(1 - \pi_i^A), \qquad d_i^B = \frac{1}{\pi_i^B}.$$

Pseudo-likelihood: final form

Step 3: Nonparametric logit model

Under the DNN model $\log \frac{\pi_i^A}{1-\pi_i^A} = g(x_i)$, we have $\log(1-\pi_i^A) = -\log\{1+\exp[g(x_i)]\}$.

The pseudo-log-likelihood simplifies to

$$\boxed{\ell^*(g) = \sum_{i \in S_A} g(x_i) - \sum_{i \in S_B} d_i^B \log\{1+\exp[g(x_i)]\}}$$

- First term: uses units from the nonprobability sample S_A.
- Second term: uses the probability sample S_B as a design-weighted proxy for the population.
- A sparse ReLU DNN approximates g_0; ADAM minimizes $-\ell^*(g)$.

DNN architecture and training

Network structure

A $(K+1)$-layer feedforward DNN with width vector $\mathbf{p} = (p_0, \ldots, p_K, p_{K+1})$:

- $p_0 = r$ (input: covariate dimension); $p_{K+1} = 1$ (output: scalar logit).
- K hidden layers with ReLU activation: $\sigma(u) = \max\{0, u\}$.
- Sparse network class $\mathcal{G}(K, s, \mathbf{p}, D)$: total number of non-zero parameters $\leq s$, sup-norm $\leq D$.

Training

- Loss function: $-\ell^*(g)$ (negative pseudo-log-likelihood).
- Optimizer: ADAM (adaptive moment estimation).
- After training: $\hat{\pi}_i^A = \{1 + \exp[-\hat{g}(x_i)]\}^{-1}$.

Proposed estimators

After training the DNN,

$$\hat{\pi}_i^A = \{1 + \exp[-\hat{g}(x_i)]\}^{-1}.$$

DNN-assisted IPW

$$\hat{\mu}_{DIPW} = \frac{1}{\hat{N}_A} \sum_{i \in S_A} \frac{1}{\hat{\pi}_i^A} y_i, \qquad \hat{N}_A = \sum_{i \in S_A} \frac{1}{\hat{\pi}_i^A}.$$

Deep doubly robust estimator

$$\hat{\mu}_{DDR} = \frac{1}{\hat{N}_A} \sum_{i \in S_A} \frac{1}{\hat{\pi}_i^A} \{y_i - m(x_i, \hat{\beta})\} + \frac{1}{\hat{N}_B} \sum_{i \in S_B} \frac{1}{\pi_i^B} m(x_i, \hat{\beta}).$$

Assumptions: identification and design

A1: Ignorability (Missing At Random)

$\pi_i^A = P(R_i = 1 \mid x_i, y_i) = P(R_i = 1 \mid x_i)$. Selection into S_A depends only on x_i, not on y_i. Hence $E(y \mid x) = E(y \mid x, R = 1)$, so the outcome model is estimable from S_A.

A2: Positivity

$\pi_i^A > 0$ for all $i \in U$. Every unit has a nonzero chance of entering S_A. Together with A1, this is the **strong ignorability** condition (Rosenbaum & Rubin, 1983).

A3: Conditional independence

$R_i \perp R_j \mid x_i, x_j$ for $i \neq j$. Membership indicators are independent across units given covariates — rules out cluster or network dependence in the selection process.

Assumptions: DNN complexity and regularity

B1: DNN architecture

The network depth K, width p_k, and sparsity s scale as

$$K = O(\log n), \qquad s = O(n\gamma_n^2 \log n), \qquad \min_k p_k \asymp \max_k p_k \asymp n.$$

Deep enough to approximate g_0, sparse enough to avoid overfitting.

B2: Smoothness of g_0

$g_0 \in \mathcal{H}(q, \boldsymbol{\beta}, \mathbf{d}, \tilde{\mathbf{d}}, M)$: a composite Hölder class. Each component has smoothness β_i and intrinsic dimension $\tilde{d}_i \leq d_i$. Sparse ReLU networks approximate such g_0 at the rate γ_n.

Assumptions: regularity conditions

Design conditions (C1, C4, C5)

- **C1**: $n_A/N \to f_A \in (0,1)$, $n_B/N \to f_B \in (0,1)$ — samples are of comparable order.
- **C4**: HT estimators are $\sqrt{n_B}$-consistent for population totals of x_i, y_i, $m(x_i,\beta)$, $m'(x_i,\beta)$.
- **C5**: $N\pi_i^A/n_A$ and $N\pi_i^B/n_B$ bounded away from 0 and ∞ — like simple random sampling.

Technical conditions (C2–C3, C6–C9)

- **C2–C3**: $m(x,\beta)$ twice continuously differentiable in β with bounded derivatives.
- **C6**: Finite moments of y_i^2, $\|x_i\|^3$ bounded; propensity-score information matrix positive definite.
- **C7**: Logistic link $\sigma(z) = (1+e^{-z})^{-1}$ has globally bounded derivatives.
- **C8**: Uniform design-consistency of HT estimators over the sparse DNN class.
- **C9**: Estimated scores truncated to $[\kappa_n, 1-\kappa_n]$ for numerical stability.

Theorem 1: Convergence rate of the DNN estimator

Setup

g_0 belongs to a composite Hölder class $\mathcal{H}(q, \boldsymbol{\beta}, \mathbf{d}, \tilde{\mathbf{d}}, M)$ where each component has smoothness β_i and intrinsic dimension $\tilde{d}_i \leq d_i$. The overall rate is $\gamma_n = \max_{i=0,\ldots,q} n^{-\tilde{\beta}_i/(2\tilde{\beta}_i+\tilde{d}_i)}$ with $\tilde{\beta}_i = \beta_i \prod_{k=i+1}^{q}(\beta_k \wedge 1)$.

Theorem 1

Under A1–A3, B1–B2, C1, C8: $\|\hat{g} - g_0\|_{L^2([0,1]^r)} = O_p(\gamma_n \log^2 n)$.

Key insight

The rate depends on the **intrinsic dimensions** $\tilde{d}_i$, not the ambient r. When $\tilde{d}_i \ll d_i$, DNNs circumvent the curse of dimensionality.

Theorem 2: Convergence of DIPW and DDR

Theorem 2

Under A1–A3, B1–B2, C1–C9 (including correct specification of g_0):

$$|\hat{\mu}_{DIPW} - \mu_y| = O_p(\gamma_n \log^2 n), \qquad |\hat{\mu}_{DDR} - \mu_y| = O_p(\gamma_n \log^2 n).$$

Interpretation

Both estimators inherit the DNN convergence rate, which depends on the composite smoothness and intrinsic dimension of g_0, not the raw covariate count.

Simulation design and main findings

- Population size: $N = 20000$; sample sizes: $n_A = 500$, $n_B = 1000$.
- True propensity score is nonlinear: interactions, squared terms, sine and log terms.
- Compared estimators: naive mean, REG, IPW, DR, DIPW, DDR.

Main patterns

- **TF**: outcome model correct, parametric propensity wrong. DIPW improves over IPW; DDR has small bias and MSE.
- **FF**: both parametric working models wrong. REG, IPW and DR deteriorate; DIPW and DDR remain more stable.

Important nuance

The strong FF performance is finite-sample evidence in this simulation, not a guarantee under arbitrary misspecification.

Empirical application

- Nonprobability data: 2015 Pew Research Center sample, $n_A = 9301$.
- Probability reference: 2015 BRFSS, $n_B = 441456$.
- Common covariates: age, gender, race, region, marital status, employment, education.
- Outcomes include neighbor trust, political participation, local voting, and drinking days.

Empirical interpretation

Adjusted estimators differ from the raw nonprobability mean, showing that the probability sample changes the target-population adjustment. Since the true population means are unknown, the application mainly illustrates implementation rather than definitive accuracy.

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